

Probabilistic Modelling and Reasoning with Python Lab

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Loan Default Prediction using R

1. Introduction

Loan default prediction plays a vital role in the finance industry. It helps lending institutions assess the credit risk of borrowers and take proactive measures to reduce financial loss. In this project, we use logistic regression to model and predict the probability that a borrower will default on a loan using features such as age, income, loan amount, and credit score.

2. Objective

The objectives of this project are:

- To simulate a dataset containing borrower details.
- To explore and visualize the distribution of features related to loan default.
- To apply logistic regression using `glm()` in R.
- To predict default probabilities and classify them based on a threshold.
- To evaluate model performance using a confusion matrix, ROC curve, and AUC.

3. Prerequisites

Install and load the following R packages:

```
install.packages(c("ggplot2", "dplyr", "caret", "pROC", "caTools"))
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
library(caret)
```

```
library(pROC)
```

```
library(caTools)
```

4. Generate Dataset

We generate a synthetic dataset of 500 records:

```
set.seed(123)

n <- 500

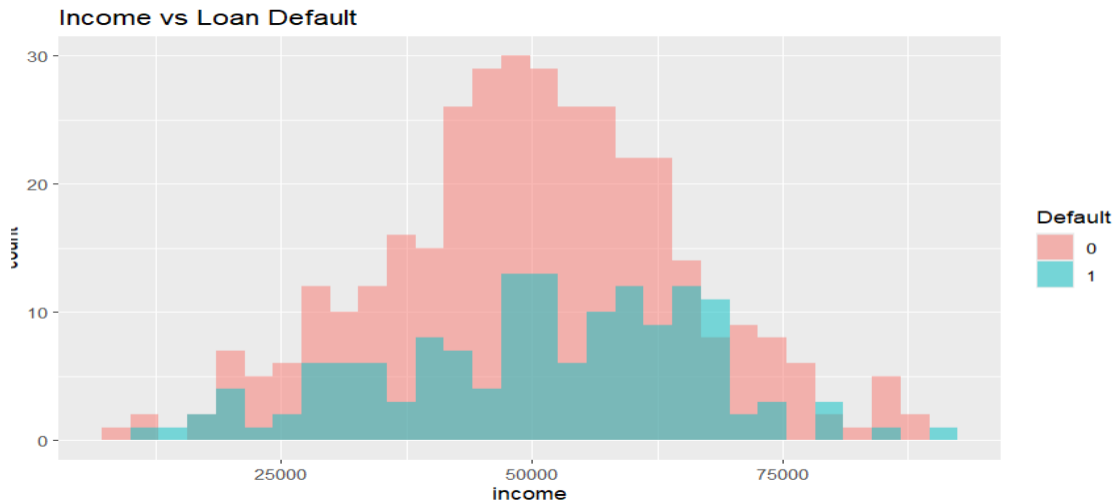
loan_data <- data.frame(
  age = round(runif(n, 18, 65)),
  income = round(rnorm(n, mean = 50000, sd = 15000)),
  loan_amount = round(runif(n, 1000, 25000)),
  credit_score = round(rnorm(n, mean = 650, sd = 70)),
  default = rbinom(n, 1, prob = 0.3)
)
```

5. Data Exploration and Visualization

Check summary and visualize default distribution:

```
summary(loan_data)

ggplot(loan_data, aes(x = income, fill = as.factor(default))) +
  geom_histogram(bins = 30, alpha = 0.6, position = "identity") +
  labs(title = "Income Distribution by Default Status", fill = "Default")
```



6. Data Splitting

Split data into training (70%) and test (30%) sets:

```
set.seed(100)
```

```
split <- sample.split(loan_data$default, SplitRatio = 0.7)
```

```
train_data <- subset(loan_data, split == TRUE)
```

```
test_data <- subset(loan_data, split == FALSE)
```

7. Logistic Regression Model

Fit a logistic regression model:

```
log_model <- glm(default ~ age + income + loan_amount + credit_score,
```

```
  data = train_data,
```

```
  family = binomial)
```

```
summary(log_model)
```

8. Predict and Classify

Generate probabilities and classify:

```
prob_pred <- predict(log_model, newdata = test_data, type = "response")  
pred_class <- ifelse(prob_pred > 0.5, 1, 0)
```

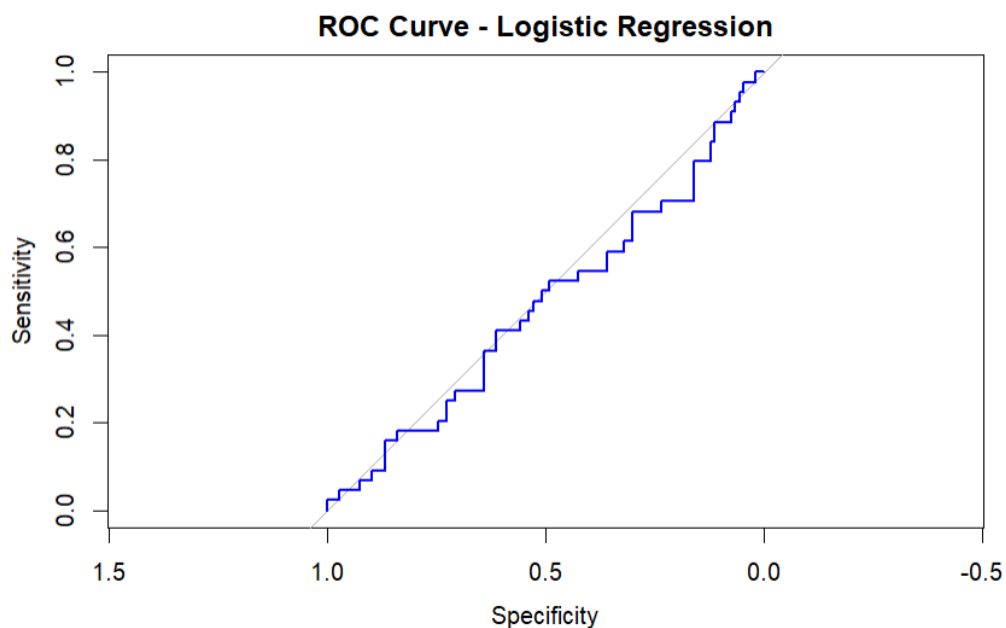
9. Model Evaluation

Confusion Matrix:

```
conf_mat <- confusionMatrix(as.factor(pred_class),  
                             as.factor(test_data$default))  
  
print(conf_mat)
```

ROC Curve and AUC:

```
roc_obj <- roc(test_data$default, prob_pred)  
  
plot(roc_obj, main = "ROC Curve - Logistic Regression", col = "blue")  
  
auc(roc_obj)
```



10. Conclusion

In this project, we successfully built a logistic regression model to predict loan default using simulated borrower data. Through data exploration and visualization, we gained insights into how features like income and credit score impact loan default.

The logistic regression model provided classification results that were evaluated using confusion matrix, ROC curve, and AUC score. These tools helped us assess the model's performance and its ability to distinguish between default and non-default cases.

This project demonstrates the practical use of logistic regression in credit risk modeling, a critical task in financial institutions. It also emphasizes the importance of model evaluation in assessing prediction reliability.

11. GitHub Repository

GitHub Link: <https://github.com/rohitraj1106/Credit-Risk-Modeling>